ELEC 221 Lecture 17
DT systems based on difference equations; the sampling theorem

Thursday 5 November 2026

Announcements

- No office hours tomorrow
- Reading break: no tutorial Monday, no class Tuesday

Last time

Continuous time:

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t} \qquad c_k = \frac{1}{T}\int_T x(t) e^{-jk\omega t}\, dt$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega) e^{j\omega t}\, d\omega \qquad X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt$$

Discrete time:

$$x[n] = \sum_{k=\langle N \rangle} c_k e^{jk\frac{2\pi}{N}n} \qquad c_k = \frac{1}{N}\sum_{n=\langle N \rangle} x[n] e^{-jk\frac{2\pi}{N}n}$$

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega}) e^{j\omega n}\, d\omega \qquad X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$h[n] = \frac{\sin(Wn)}{\pi n}$$

Last time

We evaluated the DTFT for a square pulse.

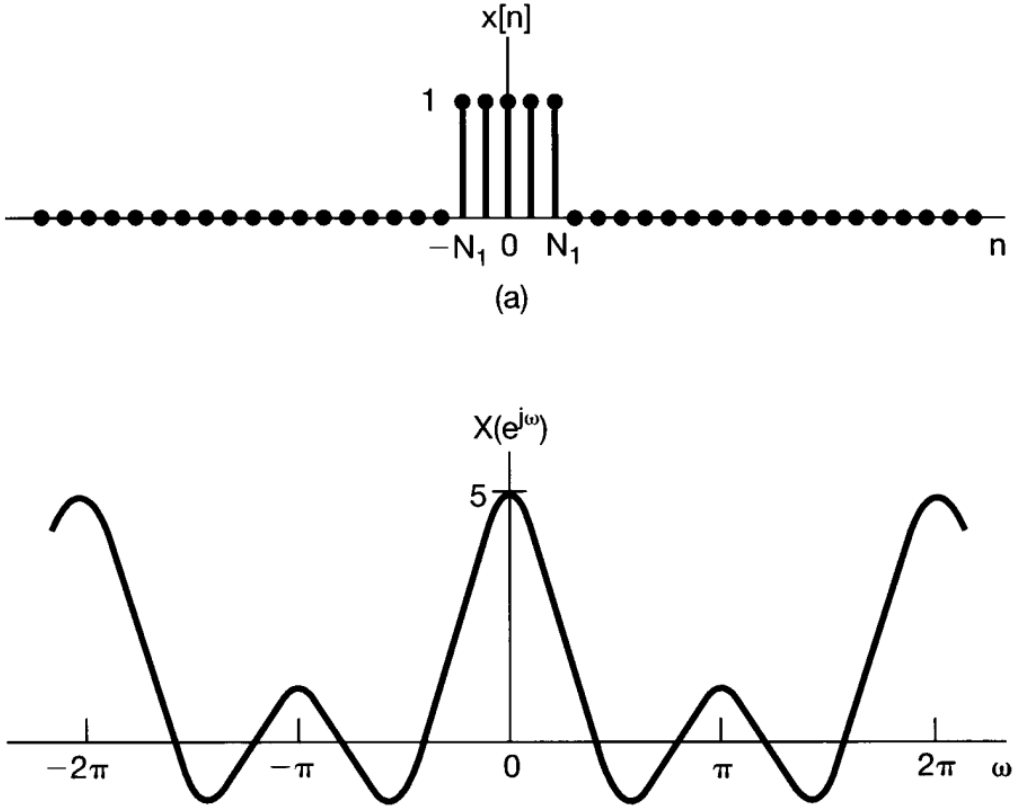


Image credit: Oppenheim chapter 5.1

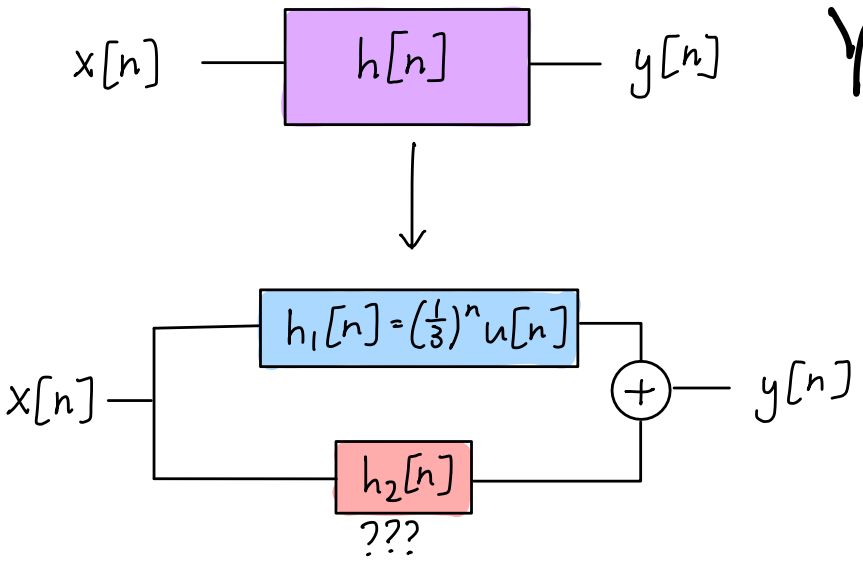
Last time

We left off with an example of applying the **convolution property** to determine the impulse response of a system.

$$y[n] = h[n] * x[n]$$

$$Y(e^{j\omega}) = H(e^{j\omega}) X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{-12 + 5e^{-j\omega}}{12 - 7e^{-j\omega} + e^{-2j\omega}}$$



Can finish solving in this week's tutorial notes

Today

Learning outcomes:

- construct and analyze DT systems based on difference equations
- state the sampling theorem
- define the Nyquist sampling rate and determine if a sampling rate is sufficient to reconstruct a signal from its samples
- describe the phenomenon of aliasing

Properties of the DT Fourier transform

$$x[n] \longleftrightarrow X(e^{j\omega})$$

Time shift:

$$x[n-1] \longleftrightarrow \sum_{n=-\infty}^{\infty} x[n-1]e^{-j\omega n} \qquad m = n-1$$

n_0

$$= \sum_{m=-\infty}^{\infty} x[m]e^{-j\omega(m+1)}$$

$$= e^{-j\omega} \sum_{m=-\infty}^{\infty} x[m]e^{-j\omega m}$$

$$= e^{-j\omega(n_0)} X(e^{j\omega})$$

Differencing:

$$x[n] - x[n-1] \longleftrightarrow (1 - e^{-j\omega}) X(e^{j\omega})$$

Example: difference equations

What is the frequency response of a system defined by a linear, constant-coefficient difference equation

$$\mathcal{F}\left(\sum_{k=0}^{N} a_k y[n-k]\right) = \mathcal{F}\left(\sum_{k=0}^{M} b_k x[n-k]\right)$$

Frequency response $H(e^{j\omega})$

linearity

$$\sum_{k=0}^{N} a_k \mathcal{F}[y[n-k]] = \sum_{k=0}^{M} b_k \mathcal{F}[x[n-k]]$$

time shifting

$$\left(\sum_{k=0}^{N} a_k e^{-j\omega k}\right) Y(e^{j\omega}) = \left(\sum_{k=0}^{M} b_k e^{-j\omega k}\right) X(e^{j\omega})$$

convolution

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^{M} b_k e^{-j\omega k}}{\sum_{k=0}^{N} a_k e^{-j\omega k}}$$

DT systems based on difference equations

amount[n] = a · amount[n-1] + deposit[n]

Exercise: compute frequency response of a system characterized by

$$y[n] - ay[n-1] = x[n]$$

$$H(e^{j\omega}) = \frac{1}{1 - a e^{-j\omega}}$$

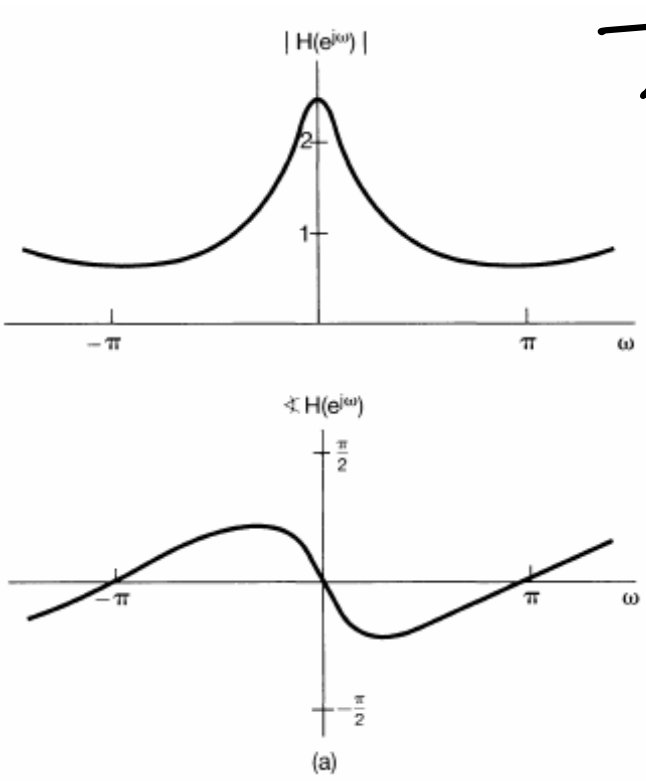
Last time: system w/ impulse response

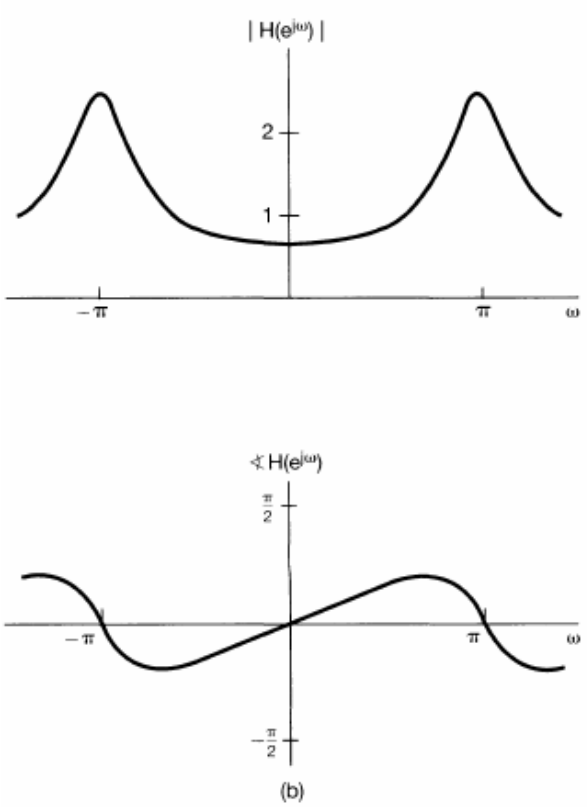
$$h[n] = a^n u[n]$$

Example

"First-order recursive DT filters"

$$\frac{1}{1-ae^{-j\omega}}$$





$0 < a < 1 \ (a = 0.6)$

$-1 < a < 0 \ (a = -0.6)$

Image credit: Oppenheim Figure 3.34

DT systems based on difference equations

Exercise: compute frequency response of a system characterized by

$$y[n] = \frac{1}{3}(x[n+1] + x[n] + x[n-1])$$

$$H(e^{j\omega}) = \frac{1}{3}\left(e^{j\omega} + 1 + e^{-j\omega}\right)$$

$$= \frac{1}{3}(1 + 2\cos\omega)$$

Can work through the general case in tutorial notes.

DT systems based on difference equations

"Non-recursive DT filters"

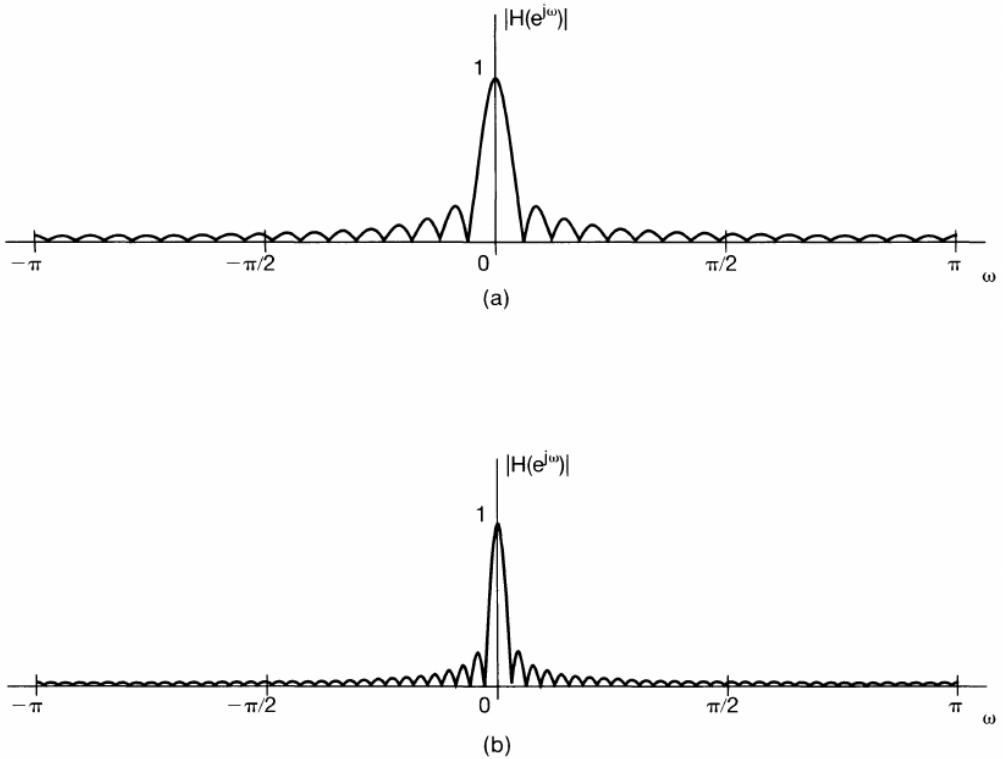


Image credit: Oppenheim Figure 3.36

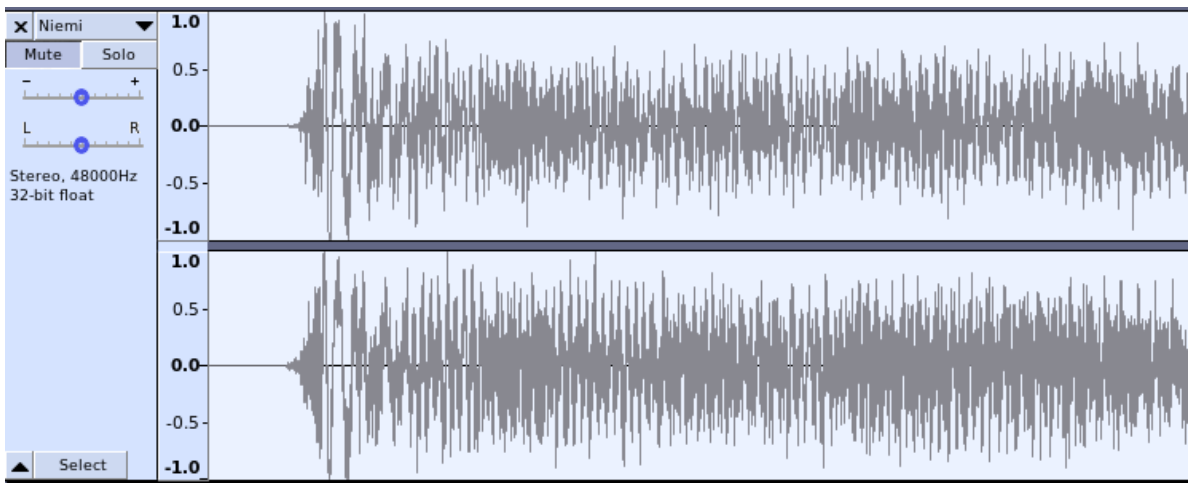
Motivation

Lecture 04 Demos

```
import numpy as np
import matplotlib.pyplot as plt
from IPython.display import Audio
```

Demo 1: fun with square waves

```
tone = 65           # A frequency in Hz
duration = 2         # The length of the audio signal (in seconds)
sample_rate = 48000  # The number of samples per second to take
t_range = np.linspace(0, duration, sample_rate * duration) # Range of time
```



Motivation

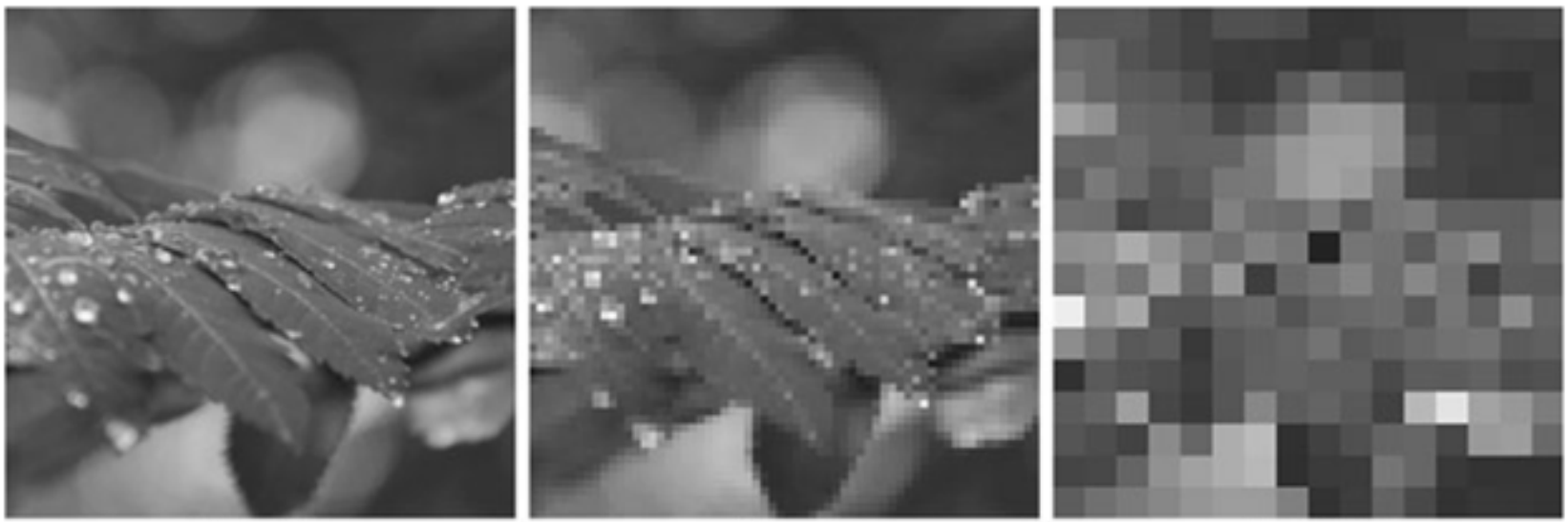


Image credit: `https://what-when-how.com/introduction-to-video-and-image-processing/image-acquisition-introduction-to-video-and-image-processing-part-2/`

Motivation

https://youtu.be/B8EMI3_0T00?t=9



History of frame rate in film:
https://www.youtube.com/watch?v=mjYjFEp9Yx0

Impulse train sampling

Core question: under what conditions can we recover a continuous time signal using only information from its samples?

The unit impulse as a sampler

Multiplying the signal by a shifted impulse picks out the value of the signal at that point:

$$x[n] \cdot \delta[n-k] = x[k] \cdot \delta[n-k]$$

This allows us to write any signal as a **superposition of weighted impulses.**

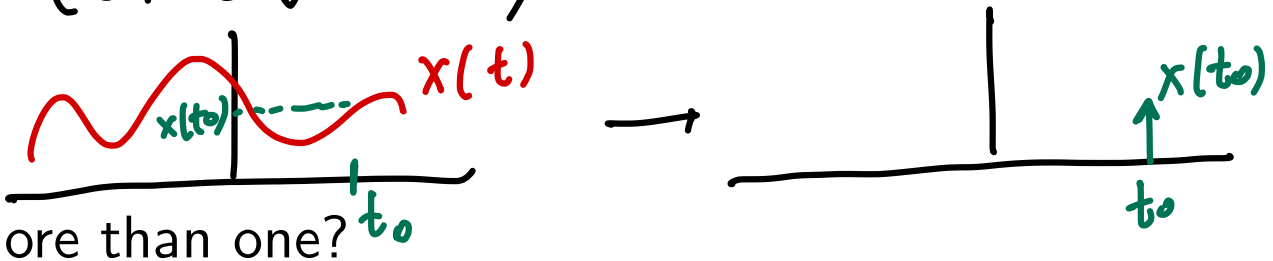
$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot \delta[n-k]$$

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Impulse train sampling

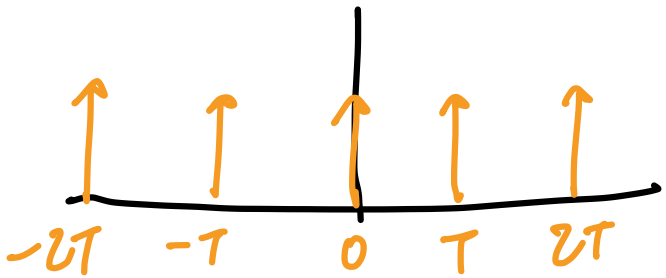
In continuous time:

$$x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0)$$



What if we have more than one?

$$p(t) = \sum_{k=-\infty}^{\infty} \delta(t-kT)$$



where

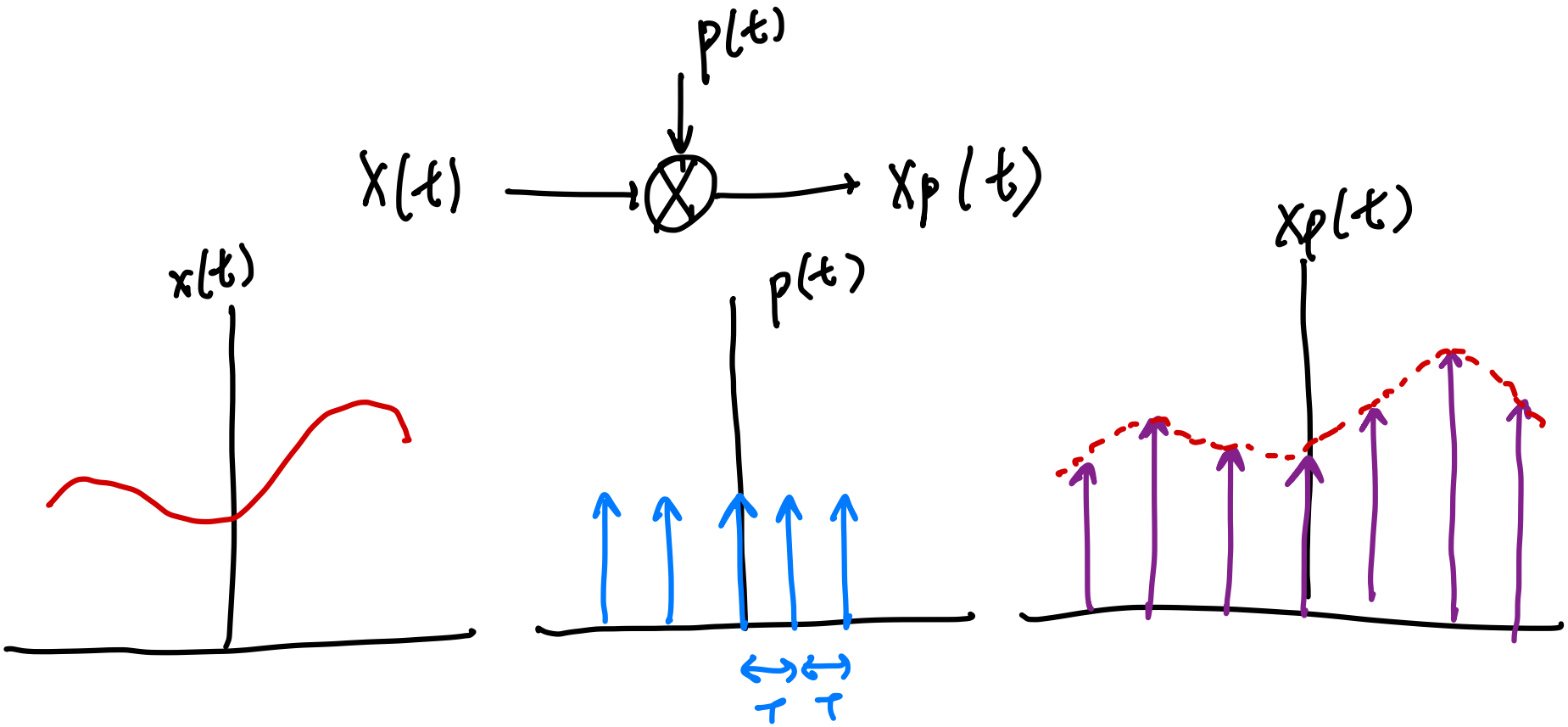
T: sampling period

$\omega_s = \frac{2\pi}{T}$ sampling frequency

Impulse train sampling

What does the following signal look like?

$$x_p(t) = x(t)\,p(t)$$



Impulse train sampling

The combined signal in the time domain is

$$x_p(t) = x(t)\,p(t) = \sum_{k=-\infty}^{\infty} x(kT)\,\delta(t-kT)$$

What happens in the *frequency domain*?

$$x_p(t) = x(t)\,p(t) \quad \longleftrightarrow \quad X_p(j\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\theta)P(j(\omega-\theta))\,d\theta$$

what is $P(j\omega)$??

Impulse train sampling

We have a periodic impulse train. Recall what Fourier transforms of periodic signals looked like:

$$x(t) = e^{jw_0 t} \longleftrightarrow X(jw) = 2\pi\delta(w - w_0)$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jkw_0 t} \longleftrightarrow X(jw) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(w - kw_0)$$

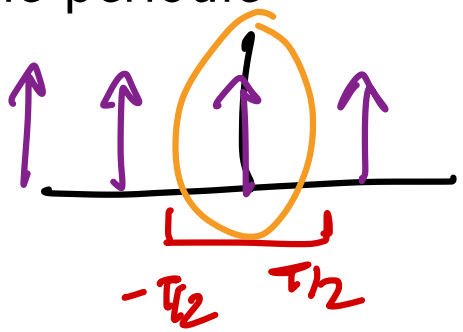
what is the Fourier series of $p(t)$?

then plug in.

Impulse train sampling

We need to find the Fourier series coefficients of the periodic impulse train.

$$p(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$



$$a_m = \frac{1}{T} \int_{-\tau/2}^{\tau/2} p(t) e^{-jm\omega_s t} \, dt$$

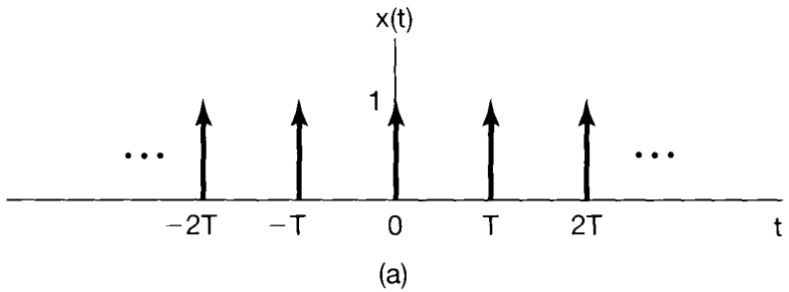
$$a_m = \frac{1}{T}$$

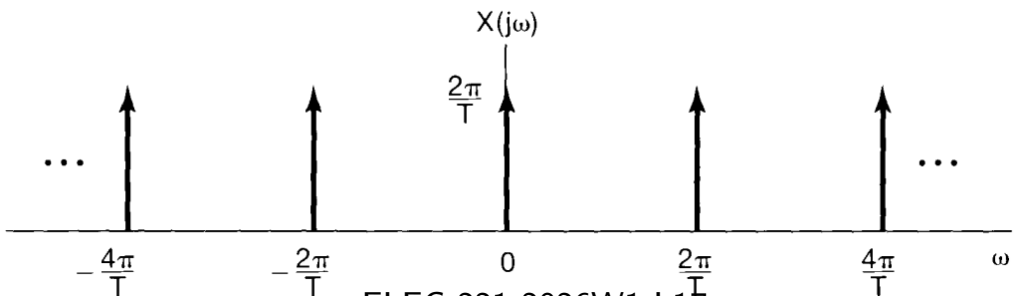
$T = \frac{2\pi}{\omega_s}$

$$P(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_s) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{T} \delta(\omega - k\omega_s)$$

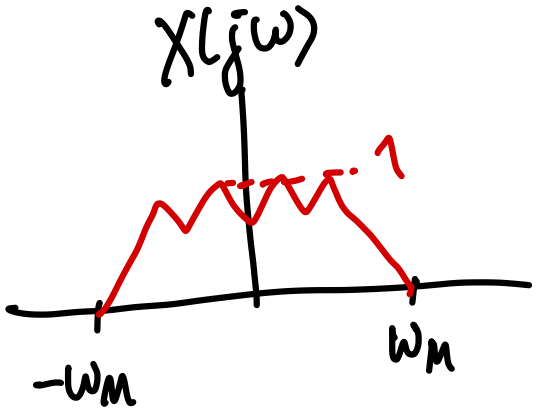
Impulse train sampling

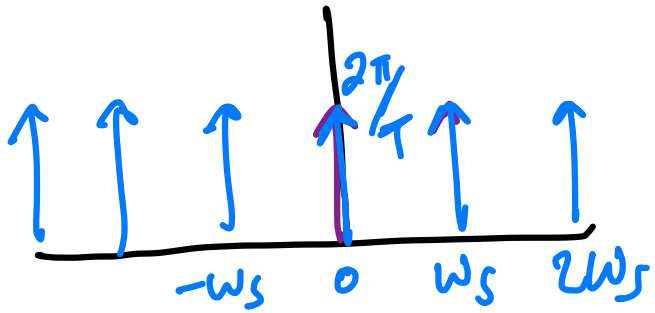
$$P(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s)$$



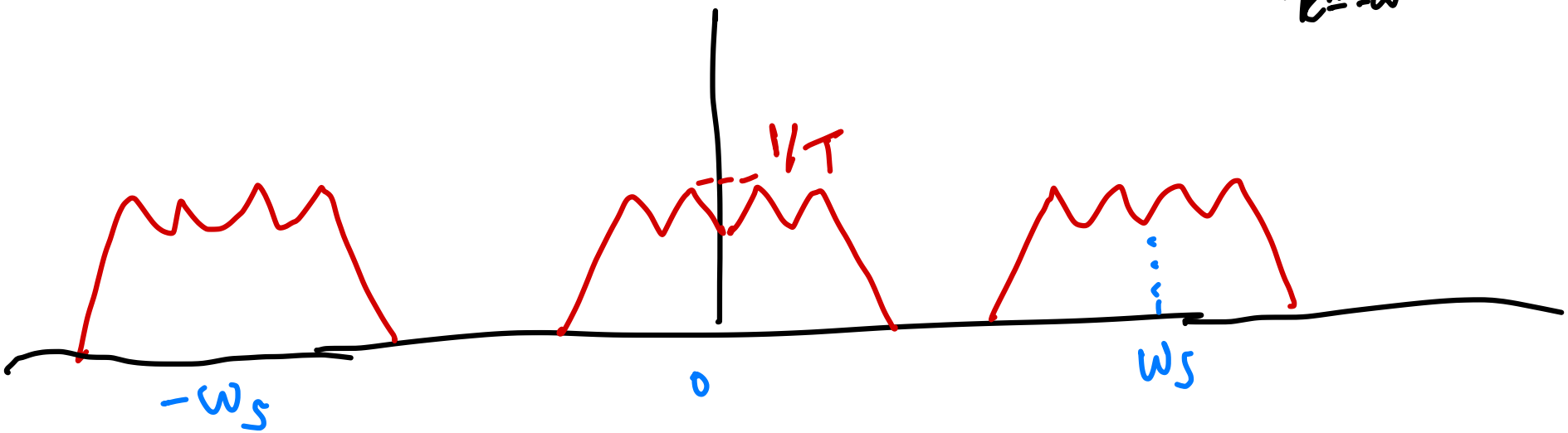


Impulse train sampling





$$x_p(j\omega) = X(j\omega) * P(j\omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))$$



Impulse train sampling

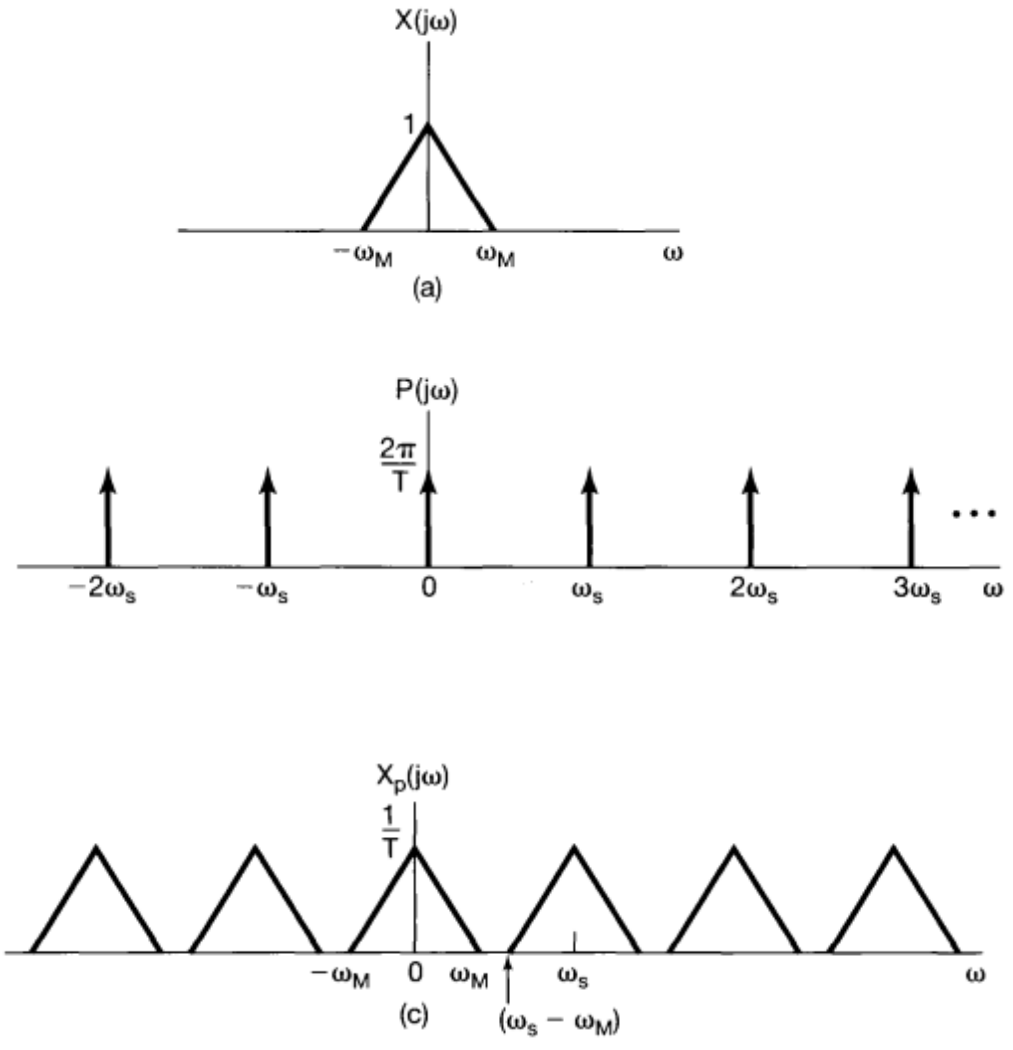
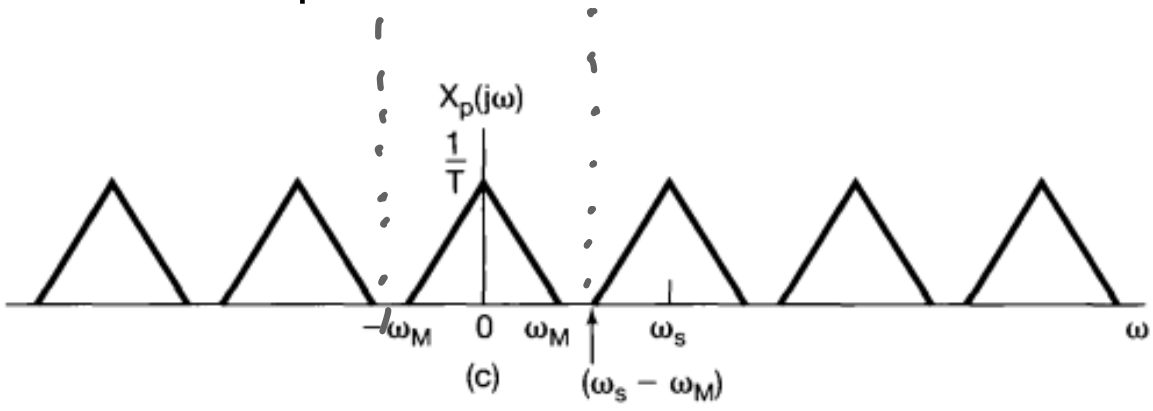


Image credit: Oppenheim 7.1

Impulse train sampling

Suppose we have sampled...



How do we recover our original signal from this spectrum?

Apply lowpass filter w/ gain T

Then: inverse FT.

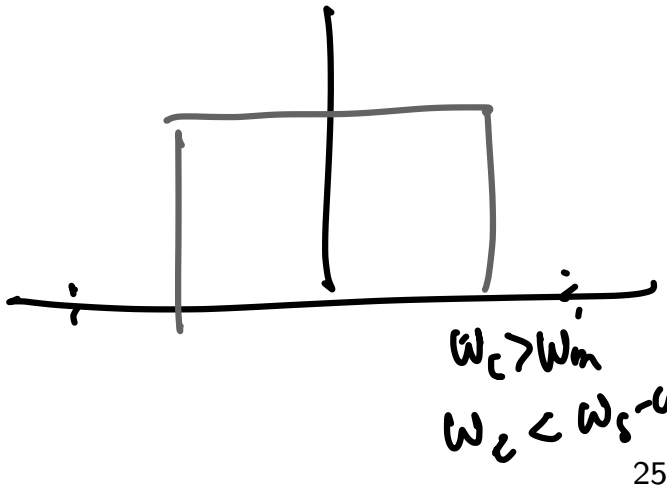
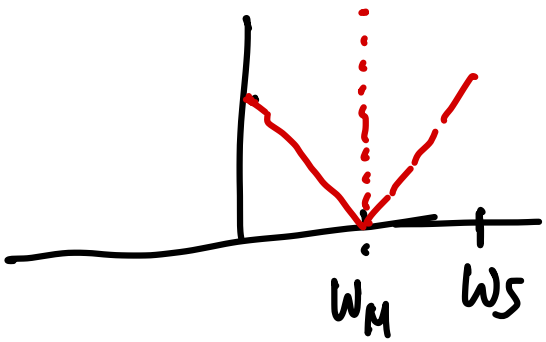


Image credit: Oppenheim 7.1

The sampling theorem

"Let $x(t)$ be a **band-limited** signal with $X(j\omega) = 0$ for $|\omega| > \omega_M$. Then $x(t)$ is uniquely determined by its samples $x(nT)$, $n = 0, \pm 1, \pm 2, \ldots$, if



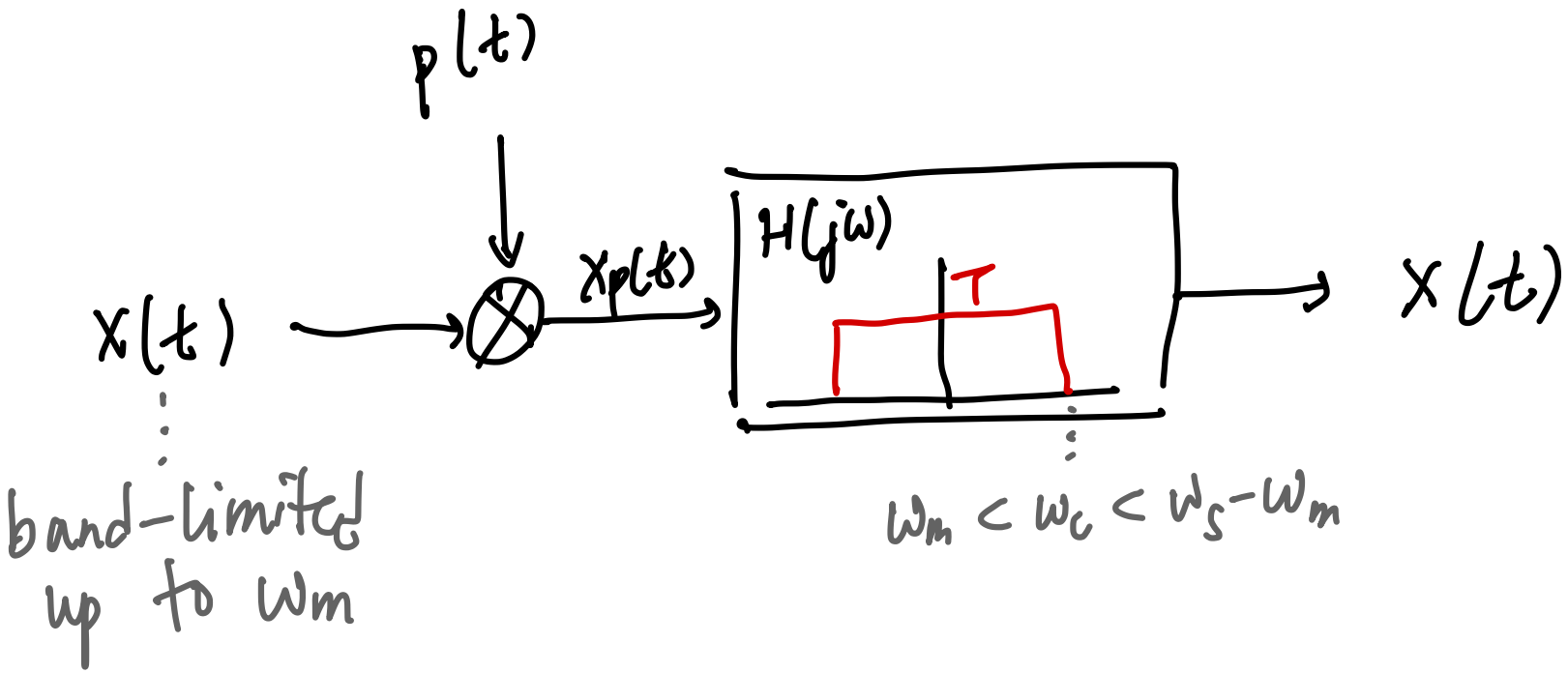
$\omega_S > 2\omega_m$

$\Rightarrow \omega_S = 2\omega_m$: Nyquist rate

Given these samples, we can reconstruct $x(t)$ by generating a periodic impulse train in which successive impulses have amplitudes that are successive sample values. This impulse train is then processed through an ideal lowpass filter with gain T and cutoff frequency greater than ω_M and less than $\omega_s - \omega_M$. The resulting output signal will exactly equal $x(t)$."

The sampling theorem

Let's show this graphically:



The sampling theorem

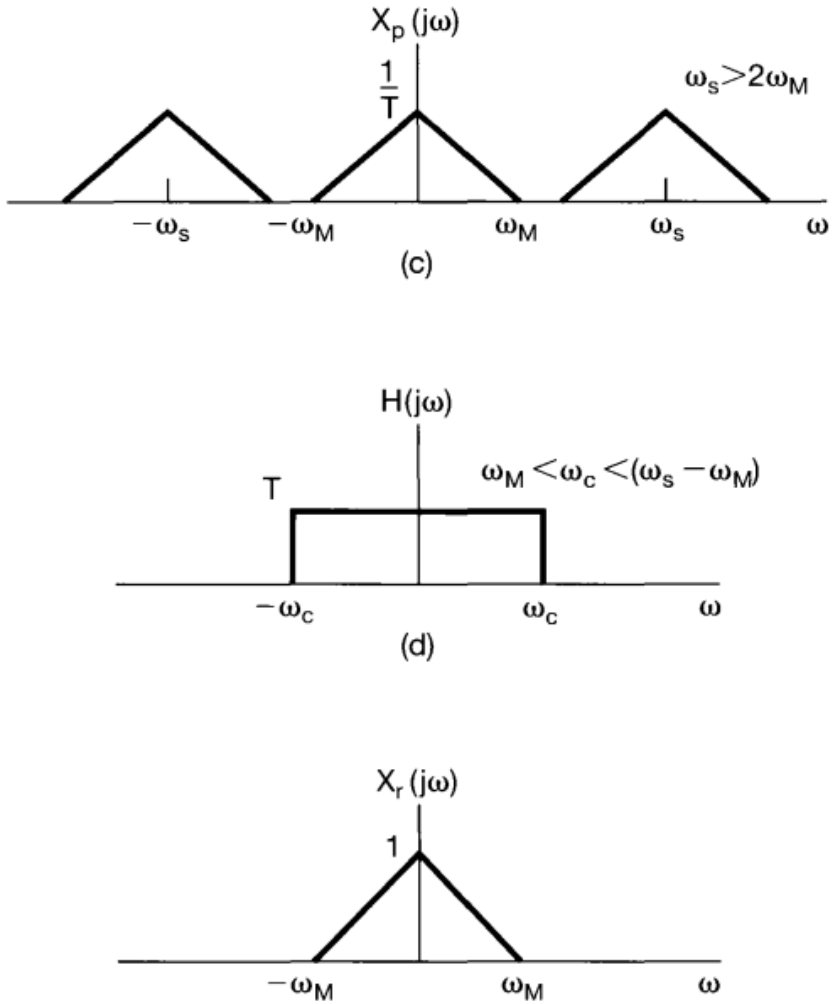


Image credit: Oppenheim 7.1

For next time

Content:

- DT processing of CT signals
- Sampling in discrete time
- Decimation/interpolation

Action items:

1. Find examples of sampling and aliasing in real life.
2. Watch TV / movies or play games at different frame rates to study the difference.
3. Enjoy the reading break!

Recommended reading:

- From this class: Oppenheim 7.0-7.3
- Suggested problems: 7.1-7.6, 7.21, 7.25
- For next class: Oppenheim 7.4-7.6